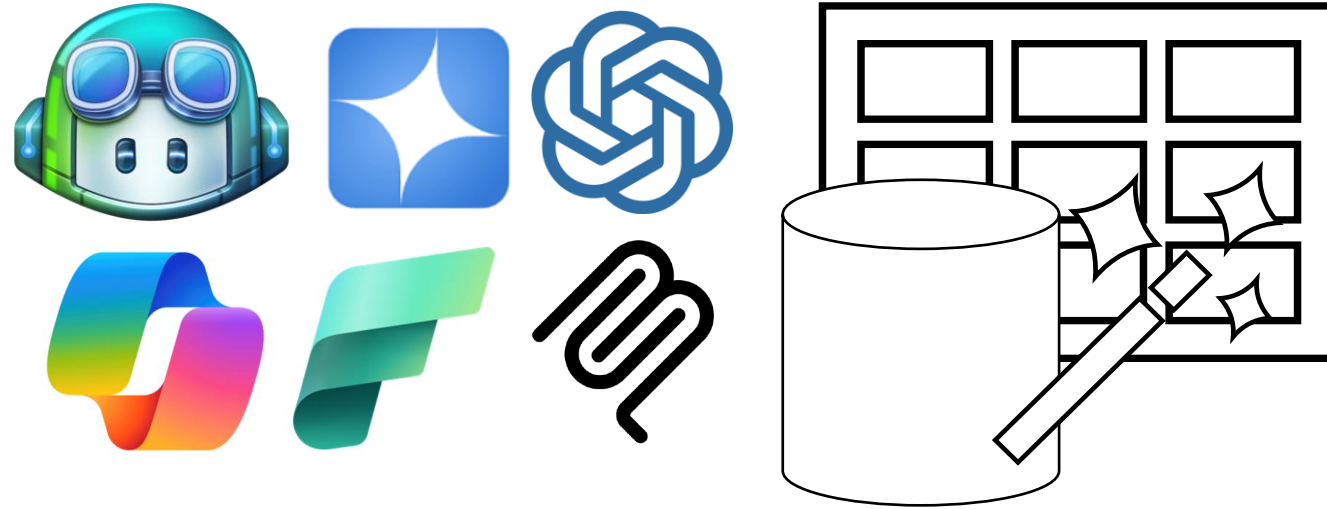




Leveraging Large Language Models with Power BI

May 23rd, 2026

About Me

Experience

- 12+ years with Microsoft BI stack
- 10+ years with Power BI
- 7+ years with Azure DevOps

Contact Me

 [John Kerski](#)

 [Blog](#)

Credentials

- MCSE: Productivity
- MS: PBI Data Analyst
- PMP



John Kerski



Slides:



Agenda

- The Challenge
- Scenarios
 - GitHub Copilot
 - MCP
 - Azure OpenAI
 - Copilot in Fabric
- Lessons Learned

The Challenge



The Challenge (cont.)



Scenario 1



Scenario

What was the table this was merged with?

What column was removed?

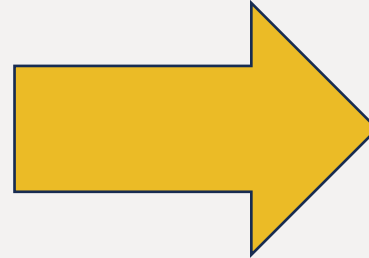
What column was added?

APPLIED STEPS

- Source
- Changed Type
- Split Column by Delimiter
- Changed Type1
- Added Custom
- Replaced Value
- Removed Columns
- Merged Queries
- Expanded DateDim
- Removed Columns1
- Renamed Columns
- Sorted Rows
- Removed Columns2
- Merged Queries1
- Expanded EyeColorDim
- Removed Columns3
- Merged Queries2
- Added Custom1
- Changed Type2
- Added Custom2
- Added Custom3
- Expanded AlignmentDim
- Renamed Columns2
- Removed Columns4
- Renamed Columns1
- Changed Type3
- ✕ Sorted Rows1

Comment Your Code

Manual Commenting Code



Prerequisites

- [GitHub Copilot](#)
- PBIP Format

GitHub Copilot Prompts

You are an assistant to help Power Query developers comment their code. Please update each line of code by performing the following:

- 1) Insert a comment above the code explaining what that piece of code is doing.
- 2) Do not start the comment with the word Step or a number
- 3) Do not copy code into the comment.
- 4) Keep the comments to a maximum of 225 characters.

You are an assistant to help Power Query developers comment their code.

Please update each line of code by performing the following:

- 1) Update the variable name explaining what that piece of code is doing.
- 2) The variable name should always start with a verb in the past tense.
- 3) The variable name should have spaces between words.
- 4) Please keep the variable to a maximum of 50 characters.
- 5) The variable name should be wrapped in double quotes and preceded by the '#'

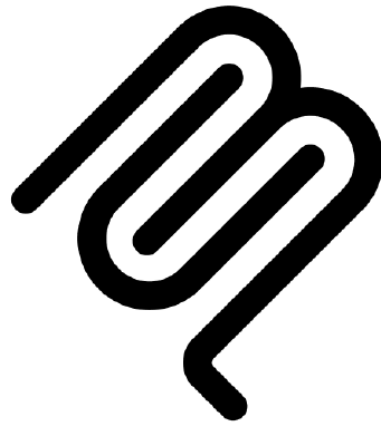
You are an assistant to help Power Query developers comment their code. Please update each line of code by performing the following:

- 1) Insert a comment above the code explaining what that piece of code is doing.
- 2) Do not start the comment with the word Step or a number
- 3) Do not copy code into the comment.
- 4) Keep the comments to a maximum of 225 characters.

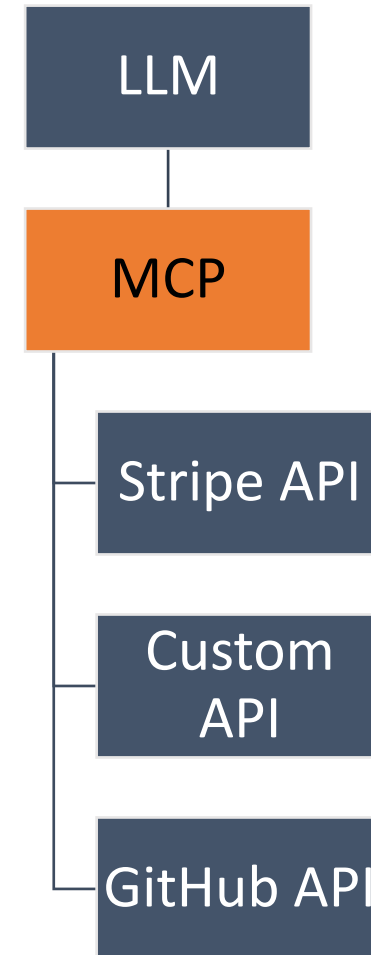
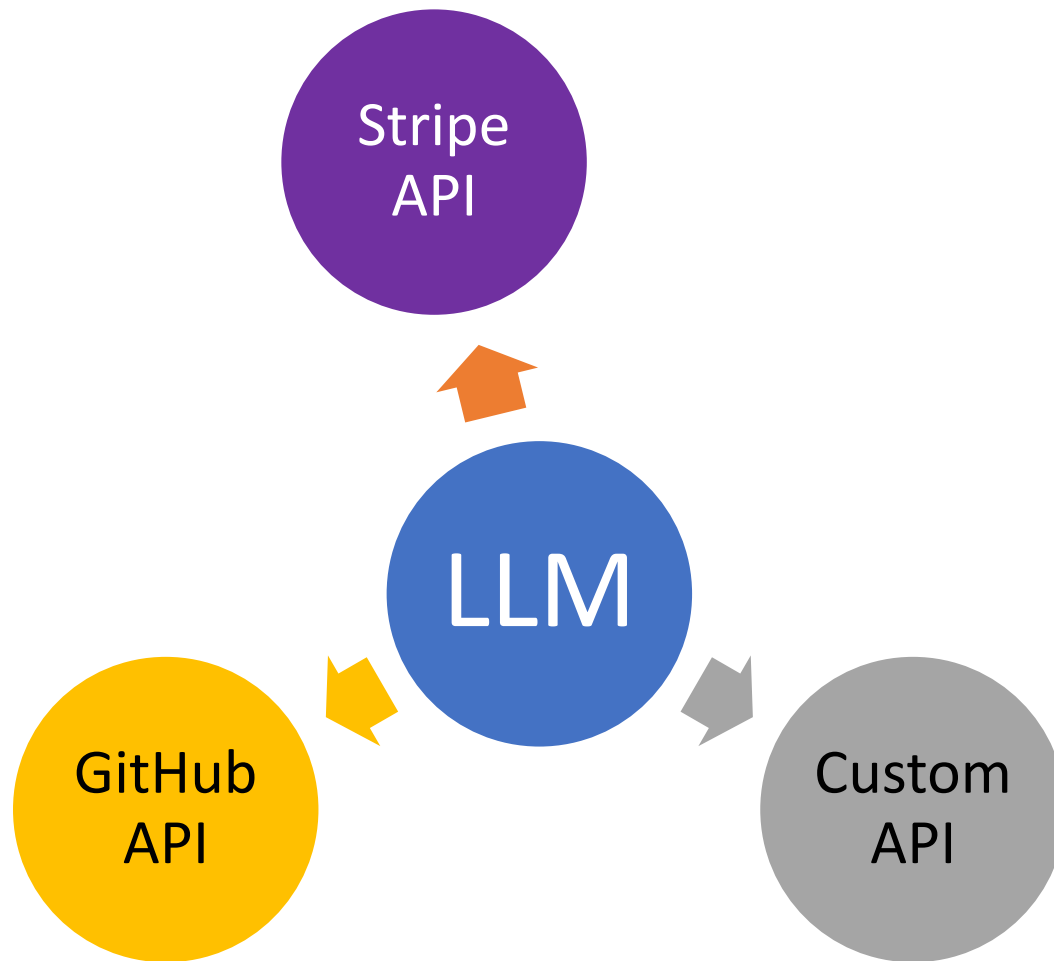
Please also update each line of code by performing the following:

- 1) Update the variable name explaining what that piece of code is doing.
- 2) The variable name should always start with a verb in the past tense.
- 3) The variable name should have spaces between words.
- 4) Please keep the variable to a maximum of 50 characters.
- 5) The variable name should be wrapped in double quotes and preceded by the '#'

Scenario 2



Model Context Protocol (MCP)



MCP

- Microsoft
 - [Power BI Modeling MCP Server](#)
 - [Introducing Fabric MCP \(Preview\) | Microsoft Fabric Blog | Microsoft Fabric](#)
 - [A Model Context Protocol \(MCP\) server implementation for Microsoft Fabric Real-Time Intelligence \(RTI\).](#)
- Community
 - [Microsoft Fabric Analytics MCP Server](#)
 - [DAX Formatter MCP Server](#)
 - [RuiRomano/powerbi-agentic-plugins](#)

Lessons Learned - MCP

- Best Practices
 - Too many tools, too much context
 - Don't trust and Do Verify
 - Have a safety net with Git (version control)
- AI-Enabled Power BI Developer
 - [YouTube Series](#)

Scenario 3



Scenario 3

POWER QUERY LACK REGULAR EXPRESSION FEATURES

- Prerequisites
 - Azure Foundry
 - A Phi-4 model deployed. Instructions can be followed [at this link](#).
 - GCC
 - [Federal-Business-Applications/whitepapers/copilot at main · microsoft/Federal-Business-Applications \(github.com\)](#)
- [Example of Using Phi 4 in Power Query](#)

Scenario 4



Scenario 4 - Prerequisites

- [Enable Copilot in Fabric](#)

Prompts

Generate a test case for each measure with the following columns:

- 1) TestName - This should describe the test conducted and be less than 255 characters
- 2) ExpectedValue - The expected value for the test.
- 3) ActualValue - The actual value returned from the DAX query.
- 4) Passed - A boolean value that compares ExpectedValue and ActualValue

The test should compare a static value with the measure to see if they match.

Generate a test case for each table that checks the table has more than zero rows using the COUNTROWS function. The output should contain following columns:

- 1) TestName - This should describe the test conducted and be less than 255 characters
- 2) ExpectedValue - The expected value for the test.
- 3) ActualValue - The actual value returned from the COUNTROWS function.
- 4) Passed - A boolean value that compares ExpectedValue and ActualValue

Lessons Learned

Lessons Learned - Prompts

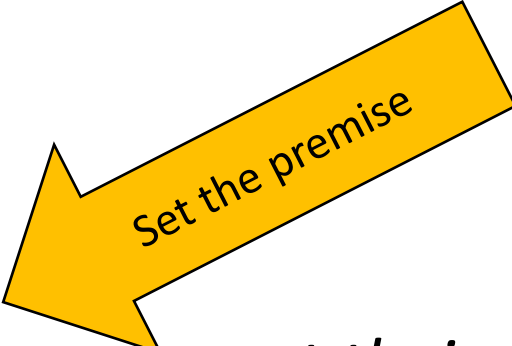
ChatGPT Prompt Engineering for Developers

You are an assistant to help Power Query developers comment their code.

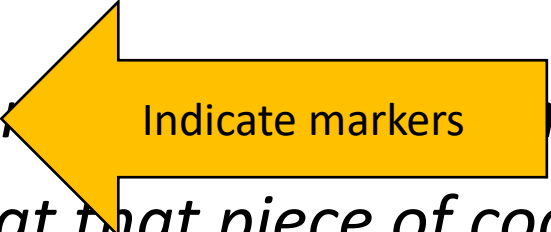
Please take the code between the `` ` ` ` in the following:

- 1) Return a comment explaining what that piece of code is doing.,*
- 2) Do not start the comment with the word Step or a number,*
- 3) Do not copy code into the comment.*
- 4) Keep the comments to a maximum of 225 characters.*

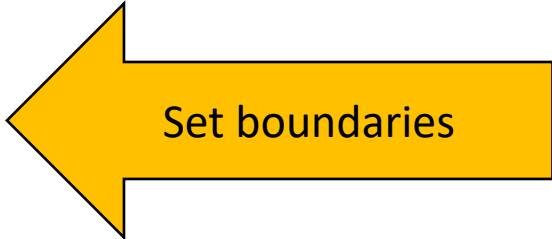
`` `[Code] ``



Set the premise



Indicate markers



Set boundaries

Security Considerations

- [Data, privacy, and security for Azure OpenAI Service](#)
- Protect the Key
 - [Vote Microsoft make this easier](#)

Questions?

